

1 Introduction

In my previous paper, I built a simple Bayesian game with a single consumer. This paper extends that setting to a small niche market with heterogeneous consumers each with a different willingness to pay. The producer does not make any parametric assumption about the distribution of consumers' willingness to pay (except regularity conditions, e.g. an atomless F with a density), but only seeks the optimal quantile that maximise the profits. This yields a general Bayesian game well-suited to a DQN. While a fully non-parametric approach can be intractable when maintaining the entire posterior over distribution, I argue that two statistics - the optimal price and the local hazard rate - characterise the producer's learning, making the model tractable.

2 Theoretical Base

In this game, we have N heterogeneous consumers, each with private willingness to pay W_i . Each consumer's policy is mixed :

$$\epsilon_i \mathbb{1}(W_i \geq x_t) + \frac{(1 - \epsilon_i)}{2}, \quad \epsilon_i \in [0, 1]$$

where ϵ_i controls the level of obfuscation. Consumers try to learn optimal ϵ_i^* via DQN.

The producer is single and updates a Bayesian belief about consumer willingness to pay. Rather than tracking the entire posterior over distributions, the producer maintains only two statistics: optimal price x^* and local hazard rate h^* at x^* . The motivation for this dimension reduction is statistical : with N consumers the producer observes an empirical CDF $\hat{F}_N$ satisfying $\sup_x |\hat{F}_N(x) - F(x)| = O_p(N^{-1/2})$ by the Dvoretzky-Kiefer-Wolfowitz inequality. Under the atomless assumption, the plug-in estimates $(\hat{x}^*, \hat{h})$ are consistent, so a market with sufficiently large N lets the producer recover the two-dimensional statistic with high probability. The producer's expected profit at price x , given aggregate obfuscation $\bar{\epsilon}$, is

$$\Pi(x; \bar{\epsilon}) = (x - c)\tilde{S}(x; \bar{\epsilon}), \quad \tilde{S}(x; \bar{\epsilon}) := \bar{\epsilon}(1 - F(x)) + \frac{1 - \bar{\epsilon}}{2}$$

where c is the unit production cost, $\tilde{S}$ is the effective demand. The constant term $\frac{1 - \bar{\epsilon}}{2}$ is a price insensitive noise floor of acceptances produced by randomisation.

2.1 Reduction of Dimensionality

Lemma 2.1. *Assume F is differentiable with density f and $\tilde{f}(x) = \bar{\epsilon}f(x) > 0$ at any interior maximiser. Then every interior optimal price x^* satisfies*

$$x^* = c + \frac{\tilde{S}(x^*; \bar{\epsilon})}{\tilde{f}(x^*)} = c + \frac{1}{\tilde{h}(x^*; \bar{\epsilon})}$$

pf. $\Pi(x; \bar{\epsilon}) = (x - c)\tilde{S}(x; \bar{\epsilon})$ is differentiable with $\Pi'(x) = \tilde{S}(x) - (x - c)\tilde{f}(x)$. At an interior maximiser $\Pi'(x^*) = 0$; dividing by $\tilde{f}(x^*) > 0$ gives $x^* - c = \tilde{S}(x^*)/\tilde{f}(x^*) = 1/\tilde{h}(x^*)$. Since $\tilde{S}(x; \bar{\epsilon}) \rightarrow \frac{1-\bar{\epsilon}}{2} > 0$ as x grows, the noise floor can push the maximiser to the upper boundary of the price domain; an interior optimum need not exist for $\bar{\epsilon} < 1$. This distortion relative to the truthful benchmark ($\bar{\epsilon} = 1$) is the mechanism by which obfuscation operates. We therefore work on a bounded price domain $x \in [0, \bar{x}]$ on which the lemma characterises any interior optimum and Kakutani gives existence.

Theorem 2.1 (Sufficient Statistic). *The producer's local optimal pricing depends on F only through the local hazard function near x^* .*

pf. By the previous lemma, any interior optimum solves $x = c + 1/\tilde{h}(x; \bar{\epsilon})$, which involves F only through $\tilde{h}(x; \bar{\epsilon}) = \bar{\epsilon}f(x)/\tilde{S}(x; \bar{\epsilon})$ evaluated near x . Hence two type distributions inducing the same effective hazard on a neighbourhood of their optimum share the same interior optimal price; the optimality check depends on F only through the local pair $(x, \tilde{h}(x; \bar{\epsilon}))$.

→the producer's learning state can be taken to be the two-dimensional $(x_t, \tilde{h}_t)$ which we use as the input to the DQN, paralleling the (μ, σ) state of the single-type model.

2.2 Existence of Equilibrium in the Game

Assume F , the CDF of consumer willingness-to-pay, is atomless. Then the expected demand $\tilde{S}$ is a continuous function of $(x, \bar{\epsilon})$ even though individual accept indicators jump - the discontinuity gets integrated out over the type distribution F . By Berge's maximum theorem, the producer's best-response correspondence $X^*(\bar{\epsilon}) = \arg\max_x \Pi(x; \bar{\epsilon})$ is nonempty-valued and u.h.c.; the induced aggregate best response map on $\bar{\epsilon} \in [0, 1]$ is u.h.c. with nonempty, convex (from mixed obfuscation strategies), compact values. Therefore by the Kakutani's fixed point theorem, the fixed point (Bayes-Nash equilibrium) exists.

3 Empirical Results

The producer observe the consumers' actions, update its belief about the acceptance curve, and reprices accordingly. Each consumers maximises their own discounted surplus, knowing producer adapts its price based to aggregate behaviour. In this game, we changed the β , the time discounting factor, to test whether patience correlates with obfuscation. As β increases, the equilibrium $\bar{\epsilon}^*$ decreases

Table 1: Equilibrium aggregate obfuscation $\bar{\epsilon}^*$ in the dynamic game with a learning producer, as a function of the consumer discount factor β (3 seeds each, $T = 25000$). For all $\beta > 0$, $\bar{\epsilon}^* < 1$: obfuscation survives against a best-responding, belief-updating producer, and decreases with patience.

β	$\bar{\epsilon}^*$ (mean $\pm$ std)
0.00	0.936 ± 0.024
0.50	0.828 ± 0.042
0.90	0.689 ± 0.077
0.99	0.717 ± 0.054

- more patient consumers obfuscate more, since discounted future gain from corrupting the producer's belief outweighs the immediate cost. The trend is monotone up to $\beta = 0.90$; the slight uptick at $\beta = 0.99$ lies within seed variation. Thus we reran the experiment with different seed settings.

Table 2: Equilibrium aggregate obfuscation $\bar{\epsilon}^*$ in the dynamic game with a learning producer, as a function of the consumer discount factor β (10 seeds each, $T = 15000$). For all $\beta > 0$, $\bar{\epsilon}^* < 1$: obfuscation survives against a best-responding, belief-updating producer, and decreases with patience up to $\beta \approx 0.9$. The Spearman rank correlation is over the raw $(\beta, \bar{\epsilon}^*)$ pairs ($n = 40$).

β	$\bar{\epsilon}^*$ (mean $\pm$ std)
0.00	0.917 ± 0.027
0.50	0.875 ± 0.068
0.90	0.788 ± 0.091
0.99	0.790 ± 0.079
Spearman $\rho = -0.630$, $p < 10^{-4}$	

With ten seeds, $\beta = 0.99$ is statistically indistinguishable from $\beta = 0.9$: the trend decreases and then plateaus rather than reversing. The overall negative correlation is highly significant (Spearman $\rho = -0.630$ with $p < 10^{-4}$). The plateau

suggests the obfuscation incentive saturates once consumers are sufficiently patient.